21 Two metal wires X and Y are connected in series and a current is flowing through them.
The number of free electrons per cubic metre in wire X is twice that in wire Y.
The diameter of the wire X is half that of wire Y.

The drift speed of the free electrons in wire X is 0.52 mm s^{-1}.

What is the drift speed of the free electrons in wire Y?

A 0.26 mm s^{-1} **B** 0.52 mm s^{-1} **C** 1.04 mm s^{-1} **D** 2.08 mm s^{-1}